

What the System Remembers: A Better Question

系统记住了什么：更好的问题

Theme: 科学、技术与互联网 · New: 45 · Review: 18 · Planned cumulative progress: 45/6, 127

English Story

A better question arrived as **a message** on **a laptop** in the middle of **a long period of development**: does this **system** **enable** a person to leave as easily as it enables them to join? The **group** had drawn up **a summary** of the **program**, but nobody had written the exit line.

The **precise measurement** of **privacy** turned out to be simple: the number of steps required to **cancel** an account. **A summary** with **numerous** footnotes, **a brilliant formulation** of the principle, **and a heap** of approvals from the **main** office were **useless** if the answer was forty-seven clicks on **a tablet**.

The false assumption was that **simplicity** is a weakness. “**Simplicity** is not the **superiority** of the lazy,” the analyst said. “It is the only thing **a slender** budget can buy.” **A vitamin** company had asked for an **attribute** they could not explain, **and a dating** service had asked for **a beverage** sponsor: both wanted data they could not justify.

The **paradox** was mathematical. In **arithmetic**, **a** zero costs nothing; in **privacy**, **a** zero is the only **permissible** value for the data you do not need. **A cardboard** box, **a torch**, **and a calculator** on the desk made the point better than the **geometry** on the whiteboard.

The **consensus** was to **minimize**, not to delete: every field would need an owner, **and** every owner would need to **fold** their data into one page. The **elegance** of the new form was that **a** user could read it aloud; the true test of **a high-tech system** is whether **a microscopic** line of **password** text can be explained by the person who typed it.

The **networking** diagram had **a shadow** of its **own**: **a fertilizer** of small features that nobody had asked for. “We cannot **solve a** trust problem with an extra field,” Maya wrote. “We can only stop adding them.” The next revision would be shorter than the previous one, **and** that would be the whole argument. It was **difficult** to see the **manipulation** in **a** form that looked so plain; the **math** was **renewable**, but an **astronomical** budget is not. The **effort** to **get** the **product** to the people who need it, to **provide** the **music** of **a** simple interface, **and** to **reach** them in **time** was the real design work.

中文翻译

一个更好的问题在一段漫长开发期的中途，以笔记本电脑上的一条消息到来：这个系统让人离开，是否和让人加入一样容易？小组拟出了程序的一份摘要，却没有人写下退出的那一行。

对隐私的精确测量原来很简单：注销一个账户需要多少步。一份带众多脚注的摘要、一条精彩的原理表述，以及总部的一堆批准，如果答案是平板上四十七次点击，那全都是无用的。

那个错误的假设是：简单是一种弱点。“简单不是懒惰者的优越，”分析师说，“它是微薄的预算唯一买得到的东西。”一家维生素公司要求一个他们解释不了的属性，一家约会服务要求一个饮料赞助商：两者都想要他们无法自证的数据。

这个悖论是数学性的。在算术里，一个零不花任何成本；在隐私里，零是你不需要的数据唯一可容许的取值。桌上一个纸板箱、一支火把和一台计算器，比白板上的几何更好地说明了这一点。

共识是最小化，而不是删除：每个字段都要有一个负责人，而每个负责人都要把自己的数据折成一页。新表格的优雅之处在于用户能把它朗读出来；一个高科技系统真正的考验，在于输入密码的那个人能否解释那一行显微镜般细小的文字。

那张网络图有它自己的影子：一堆没人要求过的小功能，像肥料一样堆着。“我们无法用多加一个字段来解决信任问题，”曼雅写道，“我们只能停止继续加。”下一次修订会比上一版更短，而这就是全部的论证。很难在一份看起来如此朴素的东西里看出那种操纵；数学是可以更新的，而一笔天文数字的预算不是。把产品送到需要它的人手上、为一个简单界面提供那种音乐感，并及时触达他们——这才是真正的设计工作。

Core Vocabulary · 核心词汇

Word / Pronunciation	Context meaning / source example
a ei POS: review required	第一个字母 A; 一个; 第一的\r; art. [计] 累加器, 加法器, 地址, 振幅, 模拟, 区域, 面积, 汇编, 组件, 异步 Collocation: — (not independently sourced) Example: n. the 1st letter of the Roman alphabet; n. the blood group whose red cells carry the A antigen
message 'mesidʒ POS: n	n. 消息, 通讯, 讯息, 教训, 预言, 广告词; vt. 通知; vi. 通报, 报告, 报信; [计] 报文; 消息; 信息 Collocation: — (not independently sourced) Example: n. a communication (usually brief) that is written or spoken or signaled; n. what a communication that is about
enable i'neɪbl POS: vt	vt. 使能够; [计] 允许, 使能, 打开 Collocation: — (not independently sourced) Example: v. render capable or able for some task
shadow 'ʃædəʊ POS: n	n. 阴影, 荫, 影子, 影像, 阴暗, 幽灵, 少许, 隐蔽处, 庇护; vt. 遮蔽, 使朦胧, 预示, 尾随; vi. 渐变, 变阴暗... Collocation: — (not independently sourced) Example: n. shade within clear boundaries; n. a premonition of something adverse; n. refuge from danger or observation; n...
solve sɒlv POS: vt	vt. 解决, 付给, 溶解; vt. 求解; vi. 求解; [计] 求解 Collocation: — (not independently sourced) Example: v. find the solution to (a problem or question) or understand the meaning of
approval ə'pru:vl POS: n	n. 赞成, 批准; [化] 核准 Collocation: — (not independently sourced) Example: n. a feeling of liking something or someone good; n. acceptance as satisfactory; n. a message expressing a favor
numerous 'nju:mərəs POS: a	a. 很多的, 数目众多的, 多数的; [机] 多数的, 甚多的 Collocation: — (not independently sourced) Example: s. amounting to a large indefinite number
brilliant 'brɪljənt POS: a	a. 光辉的, 灿烂的, 有才气的; [机] 亮的 Collocation: — (not independently sourced) Example: s. of surpassing excellence; s. characterized by grandeur; s. full of light; shining intensely
measurement 'meɪʒdmənt POS: n	n. 尺寸, 度量, 度量单位; [计] 度量, 度量单位 Collocation: — (not independently sourced) Example: n. the act or process of assigning numbers to phenomena according to a rule
summary 'sʌməri POS: n	n. 摘要, 概要; a. 摘要的, 简略的; [计] 摘要; 概要 Collocation: — (not independently sourced) Example: n. a brief statement that presents the main points in a concise form
precise pri'saɪs POS: a	a. 精确的, 严谨的, 明确的; [机] 精密的, 正确的 Collocation: — (not independently sourced) Example: a. sharply exact or accurate or delimited
math mæθ POS: n	n. 数学 Collocation: — (not independently sourced) Example: n a science (or group of related sciences) dealing with the logic of quantity and shape and arrangement
privacy 'praɪvəsi POS: n	n. 隐私, 隐居, 秘密; [计] 个人保密权 Collocation: — (not independently sourced) Example: n. the quality of being secluded from the presence or view of others; n. the condition of being concealed or h...
cancel 'kænsəl POS: n	n. 取消, 撤消, 盖销(邮票); vt. 取消, 删去, 抵销, 盖销; vi. 相互抵销; [计] 作废 Collocation: — (not independently sourced) Example: v. postpone indefinitely or annul something that was scheduled; v. make up for; v. declare null and void; make...
attribute ə'tribju:t POS: n	n. 属性, 标志, 定语; vt. 把...归于, 认为...属于; [计] 属性 Collocation: — (not independently sourced) Example: n. an abstraction belonging to or characteristic of an entity
consensus kən'sensəs POS: n	n. 合意, 一致, 同感; [计] 合意 Collocation: — (not independently sourced) Example: n. agreement in the judgment or opinion reached by a group as a whole
fold fəʊld POS: n	n. 折层, 折, 羊栏, 折痕, 信徒; vt. 折叠, 包, 合拢, 交迭; vi. 折叠起来, 彻底失败; [计] 折叠; 合并 Collocation: — (not independently sourced) Example: n. an angular or rounded shape made by folding; n. a geological process that causes a bend in a stratum of rock
vitamin 'vaɪtəmin POS: n	n. 维生素; [化] 维生素 Collocation: — (not independently sourced) Example: n. any of a group of organic substances essential in small quantities to normal metabolism
minimize 'mɪnɪmaɪz POS: vt	vt. 将...减到最少; vt. 最小化; vi. 最小化; [计] 最小化 Collocation: — (not independently sourced) Example: v. make small or insignificant; v. cause to seem less serious; play down
useless 'ju:sli:s POS: a	a. 无用的, 无效的, 无益的; [机] 无用, 无价值, 无效 Collocation: — (not independently sourced) Example: a. having no beneficial use or incapable of functioning usefully
tablet 'tæblɪt POS: n	n. 平板, 门牌, 笔记簿, 碑, 匾, 药片; [化] 片剂 Collocation: — (not independently sourced) Example: n. a slab of stone or wood suitable for bearing an inscription; n. a small flat compressed cake of some substance
torch tɔ:tʃ POS: n	n. 火把, 启发之物; [化] 火炬 Collocation: — (not independently sourced) Example: n. a light usually carried in the hand; consists of some flammable substance; v. burn maliciously, as by arson

Word / Pronunciation	Context meaning / source example
formulation ˈfɔ:mjuˈleɪʃən POS: n	n. 系统陈述, 公式化, 配制; [化] 表述; 数学表述; 数式化; 数学表述 Collocation: — (not independently sourced) Example: n. a substance prepared according to a formula; n. the style of expressing yourself
heap hi:p POS: n	n. 堆, 许多, 累积; vt. 堆积, 装满; [计] 堆 Collocation: — (not independently sourced) Example: v. bestow in large quantities; v. fill to overflow
simplicity sɪmˈplɪsɪti POS: n	n. 单纯, 简朴, 朴素, 坦率, 天真; [计] 简明性 Collocation: — (not independently sourced) Example: n. the quality of being simple or uncompounded; n. a lack of penetration or subtlety; n. absence of affectation...
manipulation mə.nɪpjuˈleɪʃən POS: n	n. 操作, 处理; [化] (用手)操作; 使用 Collocation: — (not independently sourced) Example: n. exerting shrewd or devious influence especially for one's own advantage
laptop phonetic unavailable POS: review required	[计] 膝上型的 Collocation: — (not independently sourced) Example: n. a portable computer small enough to use in your lap
fertilizer ˈfɜ:tilaɪzə POS: n	n. 肥料, 促进发展者, 受精媒介物; [化] 肥料 Collocation: — (not independently sourced) Example: n. any substance such as manure or a mixture of nitrates used to make soil more fertile
superiority sju:ˈpiəriˈbrɪti POS: n	n. 优越, 高傲; [化] 优越性 Collocation: — (not independently sourced) Example: n. the quality of being superior; n. the quality of being at a competitive advantage; n. displaying a sense of...
slender ˈslendə POS: a	a. 苗条的, 纤弱的, 微少的, 微薄的; [机] 细的, 纤细的 Collocation: — (not independently sourced) Example: s. being of delicate or slender build; s. very narrow; s. having little width in proportion to the length or he...
paradox ˈpærədɒks POS: n	n. 似非而是的论点, 自相矛盾的话, 悖论, 怪人怪事; [化] 佯谬 Collocation: — (not independently sourced) Example: n. (logic) a statement that contradicts itself
dating ˈdeɪtɪŋ POS: review required	[计] 记日期的; [经] 注明日期 Collocation: — (not independently sourced) Example: n. use of chemical analysis to estimate the age of geological specimens
beverage ˈbevərɪdʒ POS: n	n. 饮料; [化] 饮料(如牛奶、茶、咖啡、啤酒等) Collocation: — (not independently sourced) Example: n. any liquid suitable for drinking
astronomical ˌæstrəˈnɒmɪkl POS: a	a. 天文学的, 庞大的 Collocation: — (not independently sourced) Example: a relating or belonging to the science of astronomy; s inconceivably large
renewable rɪˈnju(:)əbl POS: a	a. 可更新的, 可恢复的, 可继续的, 可换新的, 可重复的, 可重说的; [计] 可更新, 可再生的 Collocation: — (not independently sourced) Example: a. that can be renewed or extended; s. capable of being renewed; replaceable
geometry dʒɪˈɒmətri POS: n	n. 几何学; [机] 几何学 Collocation: — (not independently sourced) Example: n. the pure mathematics of points and lines and curves and surfaces
cardboard ˈkɑ:dbɔ:d POS: n	n. 薄纸板; [化] 咭纸; 特等纸板; 卡纸板; 卡片纸板 Collocation: — (not independently sourced) Example: n. a stiff moderately thick paper; s. without substance
elegance ˈelɪɡəns POS: n	n. 高雅, 典雅, 优雅 Collocation: — (not independently sourced) Example: n. a refined quality of gracefulness and good taste; n. a quality of neatness and ingenious simplicity in the...
arithmetic əˈrɪθmətɪk POS: n	n. 算术, 算术知识; [机] 算术; 算术的 Collocation: — (not independently sourced) Example: n. the branch of pure mathematics dealing with the theory of numerical calculations
calculator ˈkælkjuleɪtə POS: n	n. 计算者, 计算器; [计] 计算器 Collocation: — (not independently sourced) Example: n. an expert at calculation (or at operating calculating machines); n. a small machine that is used for mathem...
networking phonetic unavailable POS: review required	[计] 连网, 建网 Collocation: — (not independently sourced) Example: v communicate with and within a group
microscopic maɪkreˈskɒpɪk POS: a	a. 显微镜的, 用显微镜可见的, 微观的, (非正式)极微小的, 细微的; [计] 微观的 Collocation: — (not independently sourced) Example: a. of or relating to or used in microscopy; s. visible under a microscope; using a microscope; s. extremely pre...
permissible pəˈmɪsəbl POS: a	a. 可允许的, 可容许程度的; [机] 容许, 允许 Collocation: — (not independently sourced) Example: a. that may be permitted especially as according to rule; s. that may be accepted or conceded
password ˈpæswɜ:d POS: n	n. 密码, 口令, 通行手段; [计] 口令 Collocation: — (not independently sourced) Example: n. a secret word or phrase known only to a restricted group
high-tech phonetic unavailable POS: n	n. 高科技 Collocation: — (not independently sourced) Example: a. resembling or making use of highly advanced technology or devices